

Vol 7 Chapter 10 — EM in Matter

Keeper (author pass — deep math/physics revision)

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Chapter 10 — EM in Matter

Level 1 — one sentence

In matter, the EM field $(\vec{D}, \vec{H})$ replaces $(\vec{E}, \vec{B})$ via constitutive relations $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ and $\vec{H} = \vec{B}/\mu_0 - \vec{M}$ where polarization $\vec{P}$ and magnetization $\vec{M}$ are the substrate's atomic K-type response to applied fields, and the frequency-dependent permittivity $\epsilon(\omega)$ and permeability $\mu(\omega)$ encode all material EM properties from glass to gold.

Level 2 — graduate-physicist precision

10.1 Macroscopic Maxwell equations

Maxwell's equations in matter (with free charges ρ_f , free currents $\vec{J}_f$):

$$\begin{aligned}\nabla \cdot \vec{D} &= \rho_f, & \nabla \times \vec{E} &= -\partial_t \vec{B} \\ \nabla \cdot \vec{B} &= 0, & \nabla \times \vec{H} &= \vec{J}_f + \partial_t \vec{D}\end{aligned}$$

with constitutive relations:

$$\vec{D} = \epsilon \vec{E} = \epsilon_0 \epsilon_r \vec{E}, \quad \vec{H} = \vec{B}/\mu = \vec{B}/(\mu_0 \mu_r)$$

For linear media, ϵ_r (relative permittivity / dielectric constant) and μ_r (relative permeability) characterize the material.

10.2 Polarization and magnetization

Polarization $\vec{P}$ = electric dipole moment per unit volume; magnetization $\vec{M}$ = magnetic dipole moment per unit volume. Linked to fields:

$$\vec{P} = \epsilon_0 \chi_e \vec{E}, \quad \vec{M} = \chi_m \vec{H}$$

with susceptibilities $\chi_e = \epsilon_r - 1$ and $\chi_m = \mu_r - 1$.

Substrate-mechanism reading: $\vec{P}$ and $\vec{M}$ are the substrate's atomic K-type response to applied $U(1)$ fields. Electric polarization shifts the atom's substrate K-type electron-cloud charge distribution; magnetization aligns substrate K-type magnetic dipole moments.

10.3 Frequency-dependent response

For time-varying fields, ϵ and μ become frequency-dependent: $\epsilon(\omega)$, $\mu(\omega)$. These are complex in general; imaginary parts represent dissipation.

Drude model for free-electron metals: $\epsilon(\omega) = 1 - \omega_p^2 / [\omega(\omega + i\gamma)]$ where $\omega_p = \sqrt{ne^2 / (\epsilon_0 m)}$ is the plasma frequency, γ is the damping rate. Explains metallic reflectivity, transparency in UV above ω_p .

Lorentz model for bound electrons (insulators): $\epsilon(\omega) = 1 + \omega_p^2 / (\omega_0^2 - \omega^2 - i\gamma\omega)$ — harmonic-oscillator response with resonance frequency ω_0 . Explains optical absorption lines, refractive-index dispersion.

10.4 Refractive index

$n(\omega) = \sqrt{\epsilon_r(\omega)\mu_r(\omega)}$ — refractive index, generally complex: $n = n' + in''$. Real part n' governs phase velocity ($v = c/n'$); imaginary part n'' governs absorption ($I = I_0 e^{-2n''\omega z/c}$).

For most materials at visible: $\mu_r \approx 1$, so $n \approx \sqrt{\epsilon_r}$.

Examples: - Vacuum: $n = 1$ - Air: $n \approx 1.0003$ - Water: $n \approx 1.33$ - Crown glass: $n \approx 1.52$ - Diamond: $n \approx 2.42$ - Negative-index metamaterials: $n < 0$ (Veselago 1968 prediction, Smith 2000 realization)

10.5 Kramers-Kronig relations

The real and imaginary parts of $\epsilon(\omega)$ are not independent — they're related by Kramers-Kronig (causality):

$$\epsilon'(\omega) - 1 = \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{\omega' \epsilon''(\omega')}{\omega'^2 - \omega^2} d\omega'$$

(with $\mathcal{P}$ denoting principal value). Measure absorption everywhere $\rightarrow$ reconstruct refractive index everywhere. Powerful constraint on material data.

Substrate reading: Kramers-Kronig is the substrate's causality requirement on Zone 1 absorption + Zone 4 emission — past inputs cause future outputs, never reverse.

10.6 Boundary conditions at interfaces

At interface between two media: $\vec{E}_\parallel$ continuous (Faraday) - $\vec{D}_\perp$ has jump = surface charge σ - $\vec{B}_\perp$ continuous (no monopole) - $\vec{H}_\parallel$ has jump = surface current

Fresnel equations for reflection/transmission amplitudes at dielectric interfaces follow from these.

10.7 Surface plasmons and metamaterials

At metal-dielectric interfaces with $\epsilon_{\text{metal}} \cdot \epsilon_{\text{dielectric}} < 0$ (frequency-dependent), the interface supports surface plasmon-polaritons — confined EM waves bound to the surface. Enable subwavelength light manipulation.

Metamaterials: artificial composites with ϵ and μ engineered to values not found in nature (negative refractive index, perfect lensing, cloaking). Substrate-mechanism: metamaterials are substrate K-type configurations designed to respond at engineered K-type-coupling rates.

10.8 BST substrate signatures in materials

Volume 9 (Condensed Matter) develops specific BST-substrate falsifiers in materials — most importantly: - BaTiO₃ 137-plane substrate eigentone (~\$25K) - Photonic crystal substrate eigentone (~\$10K) - Cuprate superconductor BST mechanism predictions - B12H32 hydride T_c ~214 K BST prediction

These are the substrate's K-type response signatures in specifically-designed materials.

10.9 Worked example: water at visible

For water at 500 nm: $n \approx 1.33$, so $\epsilon_r \approx 1.77$ (assuming $\mu_r \approx 1$). Phase velocity $v = c/n \approx 2.26 \times 10^8$ m/s.

Light enters water from air at 30° incidence. Snell's law: $\sin \theta_2 = (1/1.33) \sin(30^\circ) = 0.376$, $\theta_2 \approx 22^\circ$.

Substrate reading: at visible-frequency, water's substrate K-type configuration (H_2O molecule K-types with ~ 50 vibrational/rotational modes) gives effective $\epsilon_r \approx 1.77$. The substrate's natural mode response shifts the photon K-type wavelength by factor $1/n$.

10.10 K-audit anchors

- Chapter 3 (Electrostatics — dielectric introduction)
- Volume 9 (Condensed Matter — BaTiO₃ 137-plane, photonic crystal, cuprate)
- SP-29 / SP-30 (substrate engineering experimental program)

Level 3 — 5th-grader accessibility

EM in matter: in any material besides vacuum, the EM fields are modified by the material's response. Electric field polarizes the material ($\vec{P}$); magnetic field magnetizes it ($\vec{M}$). The macroscopic equations use $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ and $\vec{H} = \vec{B}/\mu_0 - \vec{M}$. Refractive index $n = \sqrt{\epsilon_r \mu_r}$: light slows by factor n in the medium. Water has $n \approx 1.33$; diamond has $n \approx 2.42$. Dispersion: n depends on frequency, so different colors bend differently (prism). Drude model for metals, Lorentz model for insulators — describe how electrons in matter respond to EM fields. Kramers-Kronig: causality forces a relation between absorption and refraction. BST connection: material response is the substrate's atomic K-type configurations responding to the $U(1)$ field; substrate eigentones in specifically-designed materials (BaTiO₃ 137-plane, photonic crystals) give BST falsifiers — Volume 9.

What comes next

Chapter 11 develops plasma and magnetohydrodynamics.

Where to look this up

- EM in matter: Jackson Ch 6-7; Griffiths Ch 4
- Drude/Lorentz models: Ashcroft and Mermin Ch 1, 16
- Kramers-Kronig: Jackson 7.10; Landau-Lifshitz, *Electrodynamics of Continuous Media* Ch IX
- Metamaterials: Pendry 2000; Smith et al. 2000
- BST anchors: Chapter 3; Volume 9 (Condensed Matter); SP-29/30